Abstract Vector Spaces and Subspaces

Math 5250 — Week 2

Fields

Definition. A *field* F is a set with two operations, addition and multiplication, such that:

1. Addition is associative, commutative, has an identity 0, and each x has an inverse $-x$.
2. Multiplication is associative, commutative, has an identity 1, and each nonzero x has an inverse x^{-1}.
3. Distributive law: $x(y + z) = xy + xz$ for all $x, y, z \in F$.

Examples of Fields

- The real numbers $\mathbf{R}$, the complex numbers $\mathbf{C}$, the rationals $\mathbf{Q}$.
- The two-element field $\mathbf{F}_2 = \{0, 1\}$.
- $\mathbf{F}_p$, the integers modulo a prime p.
- The field of meromorphic functions on the complex plane (or a Riemann surface).
- The field of rational functions $\mathbf{C}(T)$, or $\mathbf{C}(T_1, \ldots, T_k)$, with any field in place of $\mathbf{C}$.
- The field of formal Laurent series $F((T))$.

Vector Spaces over a Field

Definition. A vector space V over a field F is a set with:

1. Addition, satisfying associativity, commutativity, an identity 0, and inverses $-x$.
2. Scalar multiplication $x \mapsto ax$ for $a \in F$, satisfying $1x = x$, $a(bx) = (ab)x$, $a(x + y) = ax + ay$, $(a + b)x = ax + bx$.

Examples of Vector Spaces

- $\mathbf{C}^n$ and $\mathbf{R}^n$ over $\mathbf{C}$ and $\mathbf{R}$ respectively.
- $\mathbf{F}^n$, vectors of length n over any field $\mathbf{F}$.
- The space of continuous real-valued functions on $[0, 1]$.
- The space of differentiable real-valued functions on $[0, 1]$.
- The solution space of $(\partial^n + a_{n-1}\partial^{n-1} + \cdots + a_0)f = 0$, where $a_i \in \mathbf{C}$.

General Machinery

The concepts for $\mathbf{R}^n$ and $\mathbf{C}^n$ carry over to a general abstract vector space.

A map $L : V \to W$ between $\mathbf{F}$-vector spaces is *linear* if

$$L(x + y) = L(x) + L(y), \qquad L(ax) = aL(x).$$

General Machinery: Facts

1. A subspace $W \subset V$ is a subset closed under addition and scalar multiplication and containing 0.
2. Spanning sets and linear independence carry over, with coefficients in $\mathbf{F}$.
3. V is finite dimensional if it has a finite spanning set.
4. A basis is a linearly independent spanning set; all bases have the same cardinality (the dimension).
5. An $n \times m$ matrix A defines a linear map $\mathbf{F}^m \to \mathbf{F}^n$ by $x \mapsto Ax$.

General Machinery: More Facts

6. Row rank, column rank, rank, and nullity are defined as before.
7. $\mathrm{rank}(A) + \mathrm{nullity}(A) = m$, the number of columns of A.
8. A linear map $L : V \to W$ has a matrix representation $[L]_E^F$ for bases E of V, F of W.
9. The change of basis rules are the same as before.

Rank and Nullity of a Linear Map

Definition. For $f : V \to W$ linear, $\mathrm{null}(f) = \{v \in V : f(v) = 0\}$ and $\mathrm{image}(f) = \{f(v) : v \in V\} \subset W$.

Proposition. If V is finite dimensional, then

$$\dim(\mathrm{null}(f)) + \dim(\mathrm{image}(f)) = \dim(V).$$

The dimension of $\mathrm{null}(f)$ is the *nullity* of f; the dimension of $\mathrm{image}(f)$ is the *rank* of f.

Isomorphisms

Definition. A linear map $L : V \to W$ is an *isomorphism* if it is bijective (its inverse is then also linear).

If a bijective linear map $L : V \to W$ exists, V and W are *isomorphic*.

Every Finite Dimensional Space is $\mathbf{F}^n$

Let V be finite dimensional over F, with basis $E = \{v_1, \dots, v_n\}$. The map

$$L : \mathbf{F}^n \to V, \qquad (a_1, \dots, a_n) \mapsto \sum_i a_i v_i$$

is an isomorphism, with inverse $v \mapsto [v]_E$.

So every finite dimensional vector space over $\mathbf{F}$ is isomorphic to $\mathbf{F}^n$, in a way depending on the choice of basis.